

The Process to Find a Process for Governance: Nuclear Waste Management and Consent-Based Siting in the United States

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1. Introduction

“Siting storage or disposal facilities has been the most consistent and most intractable challenge for the U.S. nuclear waste management program...The erosion of trust in the federal government’s nuclear waste management program has only made this challenge more difficult.” (Blue Ribbon Commission for America’s Nuclear Future 2012, viii)

The search for a permanent repository for commercially produced nuclear waste in the U.S. remains at a standstill. Long-term objections from the state of Nevada and strong anti-nuclear sentiment from the public—stemming from distrust in historically technocratic decision-making processes—mean no commercial nuclear waste has been stored at the Yucca Mountain site in Nevada, despite its designation as sole repository in 1987. Institutional barriers to meaningful public engagement and participation mar U.S. efforts to advance a long-term management plan for spent nuclear fuel (SNF) from commercial reactors. In 2013, the Department of Energy (DOE) attempted to initiate a consent-based siting (CBS) approach to establish a novel, fair, equitable, and transparent process for managing SNF. In the wake of the 2016 U.S. election, the DOE scuttled this most current attempt at SNF management.

In this manuscript, we situate and reflect on the case of this participatory detour in attempting to manage SNF in the U.S. as part of a broader history of technocratic decision-making. We first position the U.S. CBS case amidst historical failures and limited successes of other instances of public engagement in high-level waste management decision-making in the U.S., with brief reference to international experiences. Against this backdrop, we examine the enlistment of the Expert and Citizen Assessment of Science and Technology (ECAST) network to use participatory technology assessment (pTA) as part of the federal strategy to begin creating a CBS process.

Very few entries in the public record convey the lessons from this prematurely terminated effort to create the parameters for a CBS process in the U.S.¹ Consequently, our qualitative case study (1, 2) of the U.S. consent-based siting approach against the backdrop of technocratic decision-making, combined with our critical reflection as project management participant observers within the process (3), uniquely positions us to develop and situate an understanding of the political and historical context of the complex constellation of this project. We seek to augment the public and academic records, informing future policy makers, program managers, researchers, and key stakeholders of recurring failure points to creating a CBS process. By analyzing and critically reflecting on the institutional barriers we encountered as part of this

¹ To date we are aware only of a forthcoming study by Schneider, Tomblin, and Torres, investigating “whether and how participatory technology assessments have change[d] expert cultures at NASA, NOAA, and DOE. Available at: https://www.nsf.gov/awardsearch/showAward?AWD_ID=1827826&HistoricalAwards=false.

process, we aspire to illuminate alternative ways forward for an equitable, transparent process to siting high-level nuclear waste repositories. We conclude with recommendations regarding the involvement of DOE in any future iterations of such efforts as the U.S. continues to grapple with inclusive, participatory justice in long-term decision-making for managing SNF.

1.1 Historical Context of Technocratic Nuclear Waste Management in the U.S.

Post-WWII, the clear priority for nuclear material production was linked to national security concerns. The Atomic Energy Commission (AEC) focused on producing and testing atomic weapons during the Cold War, while President Eisenhower initiated a civilian-focused privatized nuclear energy plan in the 1950s, increasing the volume of SNF waste at commercial nuclear power plants across the country (4). Scientific opinions focused on geologic repositories as the safest and most efficient method for safeguarding environmental and human health from radioactivity related to SNF (5). By the end of the 1960s, the AEC had clearly laid out its priorities for managing nuclear waste: to manage the volume and mobility of waste using natural and engineered barriers; to address transportation concerns; and to convince the public that risks to human or environmental health from nuclear materials were minor and containable (4).

Post WWII, the AEC relied on technocratic decision-making for all aspects of managing the nuclear fuel cycle by leveraging a “discourse of trust” in technological experts (6), framing nuclear energy as a key driver of American political power. Concurrently, the U.S. government technocratically recast the “destructive atom” as the “peaceful atom”(7), entrusting experts with designing technological solutions to minimize risks from radiation. This approach rendered the public as an abstract population needing to adapt this technology (8). In the post-WWII era of rebuilding national economies, atomic energy development was viewed by technocratic policymakers and nuclear proponents in countries such as France, which remains heavily reliant on nuclear power, as a way to increase energy security and independence. In this light, democratic principles of governance were subsumed under the framing of nuclear energy as a public good (9).

The late 1970s saw public attitudes in the U.S. shift away from nuclear power, transitioning toward what Blowers (6) describes as a “discourse of danger and distrust.” The period was marked by nuclear accidents, a decline in expansion of the commercial nuclear industry, and an initial accounting of the consequences on health and the environment from waste products engendered by nuclear activities ranging from weapons to energy production (6, 10). The AEC emphasized production over safety, leading to accidents and contamination at nuclear sites like Hanford, Washington (6, 11) and Rocky Flats, Colorado (12, 13). The links between nuclear energy and weapons continue to make U.S. citizens uneasy about nuclear growth (14). A growing pro-environment and anti-nuclear movement increased calls for federal transparency and accountability in the wake of the Three Mile Island nuclear incident in Pennsylvania in 1979 and the reactor meltdown at the Chernobyl plant in Ukraine in 1986 (15). In 1974, under the Energy Reorganization Act, the AEC—ill-prepared to deal with shifting public attitudes, as well as distrust from state officials—was split into two agencies. The Nuclear Regulatory Commission (NRC) was charged with licensing and regulating the commercial aspects of nuclear energy, while the Energy Research and Development Agency (ERDA) focused on the production of weapons and nuclear power (4). In 1977, the Department of Energy (DOE) was created,

tasked with creating technical specifications for a repository for SNF and a process for managing nuclear waste.

In addition to a legacy of technocratic decision-making, the DOE inherited a set of contradictory temporal, institutional, commercial, and public expectations for managing SNF. For instance, while the NRC and Environmental Protection Agency (EPA) expected DOE to create multiple test sites in different kinds of geologic media and multiple designs for engineered barriers to produce data and engender public confidence around SNF management in the near- and long-term, the commercial nuclear industry wanted a repository for SNF as soon as possible. State officials and congressional representatives also wanted the DOE to act swiftly but did not want their respective states to become repository sites. Against this backdrop, the Nuclear Waste Policy Act (NWPA) was passed in 1982 by Congress to provide clarity, funding, and a timeline for the construction of a repository (4).

Two examples in particular highlight the continuing issues with DOE's technocratic approach to nuclear waste management in the U.S. Presented briefly in the following subsections, the cases of Yucca Mountain and the Waste Isolation Pilot Plant (WIPP) exemplify pitfalls of assuming technocratic decisions can override public resistance in a democratic political system governing nuclear waste management. In the case of Yucca Mountain, nearly a half-century of technocratic and authoritarian approaches to siting continue to hamper progress toward opening a repository. In the instance of WIPP, limited success within a narrow scope was obtained after extensive negotiation with the state of New Mexico. These contrasting examples bear out Rittel and Webber's (16, p.169) observation: "To substitute expert professional judgement for those of contending political groups may make the rationales and the repercussions more explicit, but it would not necessarily make the outcomes better."

1.2 The Nuclear Waste Policy Act and the Establishment of Yucca Mountain

The Nuclear Waste Policy Act (NWPA), passed by Congress in 1982, remains the binding legal document governing disposal of high-level radioactive waste in the U.S. The NWPA established a schedule for the DOE to site, construct, and operate geologic repositories for commercially-produced SNF, and set a deadline of 1998 for the DOE to begin moving used fuel from nuclear energy facilities (17). The NWPA mandated that DOE consult with potentially affected communities and tribes, a novel situation for the DOE, which had little experience or practice in involving the public in decision-making (18). However, it also exempts the selection process for a waste site from the National Environmental Policy Act (NEPA) of 1970 that mandates public involvement in any federal decisions implicating environmental impact (8). One of the most consistent conflicts sparked by the passage of the NWPA stems from Nevadans who were neither consulted nor gave consent to bearing the long-term burden of hosting a U.S. geologic repository. A 1987 amendment to the NWPA eliminated consideration of any sites in the Eastern half of the U.S., and focused attention primarily on Yucca Mountain, gaining the bill a reputation among Nevada lawmakers as the "Screw Nevada Bill" (4). In 2002, the state of Nevada exercised their veto against the proposed Yucca Mountain repository, but a majority in Congress voted to override Nevada's veto. Many public stakeholders felt this decision rendered any attempts by the federal agencies of engaging with the public in the process meaningless and that the NWPA essentially dictated a final decision for a repository regardless of public sentiment or participatory justice (19, 4, 20).

The intractable political problem of Yucca Mountain exemplifies the top-down, technocratic approach taken by the DOE after the passage of the NHPA, which resulted in systemic participatory injustices around nuclear waste management. The DOE initially responded to public expressions of discontent with a “decide-announce-defend” model wherein federal representatives hold public meetings, describe again the goals and potential decisions under consideration, and then proceed to defend these positions in public (21, 22, 6). In such a model, the DOE retains the power to produce the scientific facts that undergird repository planning, which affects the legitimacy of public outreach that focuses on informing and educating the public, rather than engaging them in a decision-making process (23). This position has the effect of sustaining distrust amongst public stakeholders of the DOE, including local counties and Indigenous communities, who feel duly alienated from and marginalized by the decision-making process (24, 25). DOE, by fixating on technological solutions at the expense of nuanced public feedback, fell into a “complexity-exclusion trap,” by trying to reduce nuclear waste management to a technological problem, and side-stepping the more complicated social and political aspects of siting a nuclear waste repository in a democracy (26, 27, 28).

1.3 Waste Isolation Pilot Plant, New Mexico: A Study in Success?

The Waste Isolation Pilot Plant (WIPP) remains the first and only operational permanent geologic repository in the world. DOE began negotiating the WIPP site in 1972, and scientists began characterizing the site in 1975 (29). Over the subsequent 25 years, lawsuits and protests between the government, state, local activist groups, businesses, environmentalists, and Indigenous communities across the states remained locked in a political stalemate over the role of the state in relation to federal aims and goals. In the 1970s, a major point of contention arose when the DOE began discussing the possibility of using WIPP for both military and commercial waste streams without consulting the state of New Mexico (30). For scientists, the two waste streams had little technical difference, but for citizens and the state political representatives of New Mexico, this raised several concerns around the perceived overreach of the DOE, even as there was no legal requirement for the federal agency or Congress to consult with the state of New Mexico (29).

A lack of clarity remained among local, state, and federal levels regarding: the kinds of wastes permitted for disposal; how oversight was to be determined; how waste transportation would be managed; and what standards would be required at the facility to safeguard human and environmental health, as well as how involved the state of New Mexico would or could be in these decisions (31). Ultimately, New Mexico and DOE entered into a novel consultation and cooperation agreement, which helped to define the relationship among DOE, New Mexico, and Congress. In 1990, a decision was reached that WIPP would be limited to storing defense-related waste, meaning no storage of SNF would be allowed at the site (20). WIPP began receiving waste in 1999. However, these decisions, at the federal and state level, did not include direct engagement with members of the public, even as several citizen groups, both pro- and anti-WIPP, formed to challenge scientific authority and to argue for a place for the public in this process, feeding into a pattern of distrust between lay people and regulators (29). For WIPP, as well as Yucca Mountain, there were no mechanisms or processes for directly integrating the public into the decision-making process.

While resistance continues against WIPP throughout the state, some local supporters adjacent to the site feel their concerns have been heard and addressed by federal agencies, and that their local knowledge of mining and working underground has already prepared them for risks from WIPP (32). Despite two accidents in 2014, including a burst waste drum, WIPP was cleared to reopen in 2016, and began accepting waste again in 2017 (33). WIPP is still touted as a model for new geologic sites in the area. In the meantime, the DOE has filed permits to change the ways that waste is counted by volume to increase the amount of waste allowed in the site, taxing trust relationships (34). DOE also unveiled plans to keep WIPP open until 2050, instead of the previously agreed date of 2024, without fully consulting local political leaders (35). WIPP ultimately remains a warning for the intense amount of time it takes to negotiate and maintain trust among state stakeholders in a still far-from-equitable process, even though it resulted in the DOE's major goal: an actual permanent geologic repository for nuclear waste.

1.4 The Blue-Ribbon Commission and Consent-Based Siting

Yucca Mountain and WIPP are evidence of the compromised, limited effectiveness of technocratic approaches to repository siting and demonstrate clear issues with participatory justice. Participatory justice as a framework for public engagement stresses the importance of procedural attributes of decision-making, not only the outcomes. The participatory justice frame accounts for systemic institutional barriers to democratic decision-making (36) affecting public support for new nuclear projects (37). These barriers include a lack of a process for identifying, engaging, and consulting with potentially affected communities and lack of public access to timely, accurate and accessible information (18, 38, 39). Another major issue is lack of recognition of the asymmetrical power relationship between federal agencies and members of the public; a blind spot continuing to reduce trust in federal agencies (40, 24, 41). Technocratic approaches assume a lack of capacity on the part of the public to understand and therefore engage usefully on complex technoscientific issues. The technocratic assumption is of a public both uninformed and unwilling to be “correctly informed” about risks related to technoscientific projects (42).

Specific concerns related to participatory justice in nuclear waste management include: collectively demarcating who has a say in the process of determining where and how to situate SNF repositories; how to address ambiguities related to the scientific discovery process about SNF repositories; and the trustworthiness of the regulatory bodies to adhere to agreements among states, communities, and the federal government to report risks and issues. Each of these dimensions concern the sociotechnical nature of nuclear waste and underscore the complexity of nuclear waste management decisions in democratic states (39, 43, 44, 45). In addition to these systemic issues, the concept of “consent” is central to consent-based siting. Defining consent in relation to an inherently intergenerational concern presents additional challenges in defining the concept. Fundamental questions in a CBS process include how consent is gained or given, how and when it can be withdrawn, and who constitutes the community giving consent (46). Other concerns include clearly defined and collectively created understandings of risks and benefits related to such a project, and how to communicate this process to future generations (47, 39).

Under current public engagement models employed by federal agencies, many of these concerns are acknowledged but unresolved among stakeholders in the U.S. (41, 44). In part to address this fundamental lack of public trust in managing nuclear waste, the Obama administration ceased all

funding to Yucca Mountain in 2010 and assembled a Blue-Ribbon Commission on America's Nuclear Future (BRC) to reevaluate the U.S. strategy on managing nuclear waste. After two years of discussions, extensive review of national and international waste management policies and practices, and several site visits to existing nuclear waste sites and nuclear communities, the BRC published a report in 2012 with eight recommendations for DOE (49). Recommendations included creating a consent-based siting (CBS) approach that would be "[f]lexible, adaptive, and responsive" as well as "open and transparent ... with broad public participation" (49, p. 122-123).²

The BRC recommended any consent-based process be phased so decisions can be re-considered, changed, and revisited over time (50). The BRC stressed a CBS approach as the only method to date resulting in publicly-supported plans to build geologic repositories—in Canada, Finland, and Sweden—where the "technical discourse about how to safely dispose of nuclear waste is coupled to a discourse of inclusive and respectful public consultation about repository siting and criteria of safety and acceptability" (51, p. 516). Finland provides an example of a consent-based process based on a high level of institutional trust, even as other countries, such as the U.S., are undergoing a "deflation" of trust in experts (52). In Finland, communication between the power provider and the municipality was constant and consistent; the community was chosen because they already hosted two of the four Finnish nuclear plants, indicating a level of comfort with nuclear technologies. The community nearest the proposed repository site engaged in a "decision-in-principle" agreement that allowed for either government or community to withdraw from the agreement to host the site (53). The local community of Eurojoki, the Finnish state agencies involved in regulating nuclear projects, and Posiva, the power provider responsible for building the repository, negotiated a social license to operate (SLO) based on high levels of institutional trust, built over time. The approach clearly delineated roles for stakeholders in the process, as well as mechanisms for communicating risks and benefits of the repository over time in a transparent manner, with legal guarantees of veto power up to a point, and the formation of citizen groups serving in an advisory capacity (54).

In the U.S. Context, the BRC report stressed that building public trust in DOE, if possible, would be time-consuming. Moreover, if any of the BRC recommendations were to be followed, the NWPAA would need to be amended to allow for: 1) a new process and 2) a new repository site. Such a process would in theory begin addressing participatory justice concerns by being able to assess and incorporate new information from a variety of perspectives, as different management options are considered, new technologies become available, political climates shift, and lessons are learned from experiences over time. In response to the BRC, DOE pursued public engagement initiatives to find a new "process to find a process" for an approach to equitably managing nuclear waste. DOE efforts began in 2015, centered on public trust as a main concern, rather than a peripheral issue. Initial meetings formally included representatives from groups historically at the margins of these processes (including local activists and community members,

² Full list of recommendations: 1) Evaluating existing fuel cycle technologies and R&D programs, 2) interim storage, 3) permanent disposal, not limited to geologic repositories, 4) legal and commercial arrangements for nuclear fuel management, 5) decision-making process that are flexible, adaptive, and responsive, 6) open and transparent decision-making with broad public participation, 7) additional legislation or amendments to existing laws, such as the NWPAA, and 8) any other additional matters decided by the secretary of the DOE (Advisory Committee Charter, BRC Report 2012, 122-123).

environmental justice, and anti-nuclear groups, as well as scientists, policy experts and engineers). DOE convened ten such public meetings nation-wide, where expert panel presentations were followed by roundtable discussions with professional facilitators and notetakers. These public meetings were supplemented by seven additional meetings-by-request, and dozens of additional interest group meetings and conferences (17). In 2016, DOE produced a draft report based on these meetings, synthesizing public concerns around engagement and CBS in relation to nuclear waste repositories (55). In 2015, DOE also decided to pursue a second track for public engagement, which included a more in-depth and elaborate Participatory Technology Assessment approach, which we recount below.

2.The ECAST Process in a Process of Processes

While planning to conduct public meetings about the CBS process, DOE's Office of Nuclear Energy learned about the National Aeronautics and Space Administration (NASA)'s effort to use Participatory Technology Assessment (pTA) in its Asteroid Initiative project (56).³ The DOE subsequently contracted the ECAST network, of which the authors are a part, to begin engaging with public stakeholders. ECAST's efforts ultimately focused on using a pTA-informed process for determining dimensions of a CBS process. DOE's 2016 synthesis report candidly identified public trust as the largest issue needing to be addressed to create an equitable, robust, and comprehensive process for managing nuclear waste and engaging with public concerns (55). Ironically, the report makes no mention of the ECAST-led pTA (with the exception of the original agenda of the DOE Consent-Based Siting Initiative Kick-Off Meeting⁴). The secretive nature of contracting with but not publicizing the ECAST effort tracks with technocratic, opaque legacy practices of DOE SNF management; practices which present a major obstacle to integrating participatory justice into CBS approaches.

In what follows, we recount, as participant researchers, the ECAST approach to pTA as part of DOE CBS efforts. The authors include the project leader, responsible for negotiations and conference with the policy client; a senior investigator with deep expertise in U.S. nuclear energy policy and history; and a former postdoc responsible for project task execution. Materials processed to generate this manuscript include project proposal material, process documents, internal communications (e.g., emails and memos), meeting notes, and works-in-progress completed up to the point of project termination by DOE. To develop this critical case reflection, we met as authors on multiple occasions to debrief and validate individual and shared perceptions on the way pivotal events unfolded, especially in comparison to other pTA projects with federal agencies in the U.S. (57).

³ The Expert and Citizen Assessment of Science and Technology (ECAST), led by Arizona State University, is a network of university, informal science education institutions (e.g., science museums), and policy research organizations. ECAST creates opportunities for citizens to learn about an issue and share their individual views so science and technology policy makers may better understand and take into account citizens' views on how scientific and technological developments are perceived, valued, experienced, shaped by, and related to our everyday lives. Launched in 2010, ECAST's recent pTA projects include climate intervention research, automated vehicles, and gene editing (see <https://cspo.org/areas-of-focus/pta/>).

⁴ Available at:

<https://static1.squarespace.com/static/55c4c892e4b0d1ec35bc5efb/t/56969f08a2bab8c0d989a062/1452711689235/2016.01.20+-+Agenda+-+CBS+Kick-Off+Meeting.pdf>

Results are presented in two parts. First, we present the substantive outcomes of materials created with and delivered to DOE as part of the work of preparing CBS deliberations for SNF management. Second, we reflect procedurally on what it meant to manage the project amidst the technocratic culture and historical and political context of U.S. SNF management efforts (3).

2.1 The ECAST approach to pTA

The ECAST model for pTA emerged as a response to three critical and persistent failures in contemporary science policy and decision-making, namely failures to incorporate public values (58, 59, 60), to manage socio-technical uncertainties in emerging technologies (60, 61), and to address democratic gaps (63, 64, 65). Much has been studied and written on each of these failures in overlapping but separate spheres of social science scholarship and practice. Collectively, these failures contributed to the conceptualization of the engagement component of an Anticipatory Governance framework for emerging technologies, operationalized at the National Science Foundation (NSF)-supported Center for Nanotechnology and Society (CNS) at Arizona State University (66). CNS convened the National Citizens' Technology Forum on Nanotechnologies and Human Enhancement (67), adapting the Danish Board of Technology's consensus conference model (68) to convene simultaneous citizen dialogues in six U.S. locations.

These developments paralleled a similar evolution in the scholarship and practice surrounding the field of public understanding of science, where dialogue and bi-directional communication under the guise of a reconceptualized public engagement in science (PES) emerged as an alternative to the knowledge deficit model (69) and led to the creation of the Nanoscale Informal Science Education Network (NISENet), also with NSF support. The ECAST network was launched in 2010 by the leadership of CNS and NISENet respectively at Arizona State University and Boston Museum of Science in collaboration with other proponents of pTA at the Loka Institute, SciStarter, and the Woodrow Wilson International Center for Scholars. ECAST presented a convergence of all of these movements in the form of a reinvigorated participatory technology assessment (pTA) method with the explicit and declared intent to combine the social studies of science with informal science education and policy analysis in the pursuit of five operational objectives: (1) combine participation and expertise, (2) adopt an agile 21st century structure, (3) continually innovate concepts and practices, (4) be non-partisan in structure and governance, and (5) be committed to transparent process and public results (70).

The Danish Board of Technology's World Wide Views on Biodiversity provided ECAST the first opportunity to operationalize parts of this distributed model of pTA through a demonstration project. In 2012, it convened citizen deliberation forums in Boston, Washington, Denver, and Phoenix to provide input to the United Nations Convention on Biological Diversity (70). This effort provided a proof of concept for the NASA Asteroid Initiative and subsequent projects on community resilience, climate and energy, solar geoengineering, and automated mobility, which collectively contributed towards the development of a reflexive pTA method to democratize science policy decision-making (71). The earlier group of projects brought ECAST to the DOE's attention, leading its involvement in the CBS project.

2.2 ECAST, DOE, and Progress Toward a CBS Approach

Among the three systemic failures noted as providing the conceptual underpinning of the ECAST method and its evolution, DOE's interest in the approach, at least initially, was motivated by a recognition of the democratic failures of SNF management. With added familiarity of the theoretical constructs driving pTA, DOE and ECAST found an even stronger rationale in the expert recommendations for the co-management of technical and societal uncertainties (72). However, the third systemic failure, the failure to integrate public values with scientific and economic values, was not an explicit concern for DOE. With ECAST, the DOE was fundamentally "seeking the social acceptance of a technically rational choice" rather than "negotiating the technical feasibility of a societal choice" (73).

The ECAST pTA process consists of three main steps: 1) problem framing, 2) peer-to-peer discourse through public forums, and 3) results integration (71). The problem framing step is the process of negotiating the design parameters for the pTA. Informed by the state of knowledge, expertise and experience, it aims to answer questions as to who should be engaged and how, what questions need to be asked, what knowledge and information are needed to answer them, and how answers should be collected to make the results actionable. Public forums are designed within these parameters to the extent possible, depending on resource, time, legal, and budgetary constraints. The forums generally observe four guiding principles: (a) informed participants, (b) diverse representation of background, perspectives, and experiences, (c) respectful bi-directional conversation, and (d) clear and usable results. Results are gathered using various data collection instruments such as pre- and post-surveys, written individual and group responses, table observations and participant interviews. These serve as input to the third and final step wherein preliminary results are collectively reviewed by experts and stakeholders to create usable and actionable outputs for policy and decision-makers.

For the DOE CBS process, ECAST completed the problem framing step, which included brokering a pre-deliberation expert-stakeholder workshop to identify and frame issues for public discussion of SNF. However, in January 2017, during the development of the public forums in step 2, the DOE terminated the ECAST contract, citing a shift in priorities of an incoming presidential administration. The forum design process in general, and the expert and stakeholder workshop in particular, produced informative insights for future CBS initiatives that might center democratic decision-making on science and technology issues, despite the truncation of the ECAST pTA process.

During the problem framing stage of the ECAST effort, we completed a review of the relevant literature, a scoping meeting with ECAST and DOE principals, and a pre-deliberation forum design workshop with expert stakeholders. The goal of the design workshop was to engage a diverse group of expert stakeholders about the needs, objectives, and design principles for citizen deliberation forums on SNF management. We also collected advice on salient issues, questions, and relevant content for use in citizen deliberations to ensure the eventual outputs had a higher chance of aligning with DOE's decision-support needs. The ECAST team, working with the

DOE Office of Nuclear Energy, structured the expert and stakeholder workshop around the following forum design objectives:

- Desirable, problematic, and alternative ways of framing CBS issues;
- Ethical, legal and societal questions raised by the CBS approach that require informed public deliberation;
- Critical, accurate, and balanced background information related to these questions so that public participants can engage in informed deliberation.

The list of workshop invitees was co-developed by DOE and ECAST based on the literature review to ensure adequate representations of diverse perspectives and expertise in the issue domain. This task had to overcome initial hesitancy from experts and issue advocates historically skeptical of DOE's intention and commitments to a fair process. Stakeholders were invited based on their organization's historical engagement with nuclear waste management, and represented rural, urban, tribal, academic, industry, and policy interests and concerns. During one and a half days at the Museum of Science in Boston, ECAST team members engaged 14 external experts and stakeholders and eight DOE participants to discuss and deliberate on these issues.⁵

The ECAST team turned to expert and stakeholder participants for input on questions ranging from the normative to specific content questions: *Who do we target for deliberations? How do we frame the SNF problem? What questions do we ask? What context and information do we need to provide? What resources can we leverage to do this work? What are we missing from the material presented and discussed?* During the event, after introductory presentations on the process, specific short talks were invited by attending experts and stakeholders, followed by a series of small-group conversations in which issues and probing questions were refined.

Following the workshop, ECAST compiled and drafted an initial deliberation plan to circulate for review with a subset of workshop attendees. We structured the deliberation plan on the premise that well-informed public discussion is driven by citizens' awareness of multiple values involved in an issue, as well as the historical context of why and how a particular deliberation is important (74). In this case, we had to find a delicate balance between contextualizing the problematic history of public engagement and systemic distrust in federal management of SNF, while also recognizing that some publics are more accepting and supportive of SNF discussions than others (75). The team proceeded to design the public deliberation forums in concert with extended expert and DOE review, as well as to compile background material to prepare for public participants, up until the early termination of the project.

⁵ Attendees included: Allison Fisher, Public Citizen; Andrew Kadak, Nuclear Economics Consulting Group (NECG) Affiliate; Beatrice Brailsford, Snake River Alliance; Brian Gutherman, Gutherman Technical Services; Christopher Recchia, State of Vermont; David Ropeik, Harvard Extension; Don Hancock, Southwest Research and Information Center; Glenn Paulson, George Washington University; Jen Schneider, Boise State University; Kara Colton, Energy Communities Alliance (ECA); Phil Niedzielski-Eichner, Governmental Dynamics; Phil Richardson, Galson Sciences Ltd; Rod McCullum, Nuclear Energy Institute; Seth Tuler, Worcester Polytechnic Institute.

3. Results from ECAST pTA Efforts with DOE

Below, we offer two separate sets of results from the work completed: first, brief substantive findings about CBS deliberations for SNF management; second, procedural findings and discussion of the institutional barriers and deep-rooted technocratic culture of DOE that curtailed a pTA to developing the CBS process.

3.1: Substantive Results on SNF Management and CBS

Examples of equity, trust, and uncertainty questions initially reviewed by the expert and stakeholder team are shared in Table 1. Normative themes raised by the BRC as well as agencies like the DOE included values affecting equity (justice, fairness, morals, ethics); trust (transparency, culture of safety, nature of expertise); and uncertainty (flexibility; changes in certainty). Content issues in the workshop focused on how to define considerations around consent, including: defining the term, parties to consent, the nature of consent, and the content of consent, including when and how consent can be given and withdrawn.

Table 1: Normative themes and potential deliberative questions reviewed at the issue-scoping workshop. Each table was staffed with at least two ECAST members, one to serve as facilitator and one as notetaker to ensure dialogue and perspectives were supported and recorded.

Normative theme	Example deliberative questions
Equity	What would be required to ensure that costs and benefits are distributed equitably among <i>present</i> parties to consent? What would be required to ensure that costs and benefits are distributed equitably among <i>future</i> parties implicated by consent (i.e., future generations)? Will disadvantaged communities be more likely to serve as a target in this question? Why? How can that be avoided? Are there other equity issues to consider that we have missed?
Trust	What would a transparent process for sharing information about this issue have to include? What would a transparent process for responding to concerns about this issue have to include? What might independent oversight look like related to supporting engagement with/resolution of these issues? Are there other trust issues to consider that we have missed?
Uncertainty	What processes need to be in place to allow for changes to the makeup and organization of participants? What processes need to be in place to allow for changes to knowledge? What processes need to be in place to allow for political changes?

After the workshop, these normative themes were re-framed as guiding principles to be collectively addressed and defined through any future CBS process. While many of these themes had been previously identified as historical points of persistent contestation in SNF management, they had not been substantially defined as points needing to be elaborated upon through public engagement in relation to a legitimate and equitable CBS process. Using the foundation of the framing workshop, the ECAST group developed working definitions for these guiding principles as starting points for the larger public deliberations. These guiding principles consisted of:

- 1) **Equity:** ensuring fairness and justice related to procedures and outcomes;
- 2) **Trust:** committing to transparency, safety, and empowerment of host communities that process will be honored and kept by all parties over time;
- 3) **Well-being:** addressing community and environmental health and economic development needs;
- 4) **Power:** negotiating and distributing authority among different levels of decision-making, including the right to withdraw consent; and
- 5) **Longevity:** flexibility and adaptability to changing circumstances over time due to changing societal needs and values.

Beyond these terms, the concept of consent was also a major concern for workshop participants. In general, the ECAST framing workshop helped to elaborate how consent processes benefit when:

- The concept of consent is established before decisions are made.
- Consent can be given freely, in ways where power in planning, decision making, and bargaining is equal, and where there is transparency and no manipulation or undue leverage compromising consent.
- There are points where consent can be withdrawn, understood, and upheld by all parties as the process is supported with legal channels for resolving issues as they arise.
- Communities are empowered with easy-to-access, easy-to-understand, independently verified information.
- The process has ample opportunities for people to ask questions as they learn.
- The process respects cultural practices and the ability of people in a community to make choices on their own behalf.

After the completion of the problem framing step, ECAST began planning for the second step, the peer-to-peer through public forums.

3.2 Institutional Barriers to a pTA Approach for CBS Process Development

A range of institutional barriers to a pTA approach to informing and developing a CBS process became apparent during the ECAST work. We describe the cultures, practices, and rules adopted by DOE, drawing specifically on the Institutional Analysis and Development Framework for our elaboration (76, 77). We do not carry out a strict institutional analysis of nuclear waste management governance, as this would be beyond the scope and intent of the present manuscript. More simply, we observe key ways the attributes of DOE and its legacy contribute to institutional barriers to a CBS process, based on our experience and working knowledge of the case and theory.

The institutional barriers observed are grounded in the historically technocratic culture of decision-making used by the DOE. A CBS process foregrounds the right of the public to

question aspects of SNF and HLW management. This right has been closely and vehemently guarded or stonewalled by technocratic approaches preceding and continued by DOE. In this context, questions elaborating who should be involved in a CBS process; at what point in the decision-making process; how should this be decided; and even what defines a “local community” might be perceived not only as threats to DOE but also to Congressional authority. Beyond questions of “who,” a CBS raises questions of “how” a CBS would work—what *are* appropriate ways people should be supported to learn about sociotechnical aspects of nuclear waste management *before* even raising ideas of consent? How *should* “informed consent” be defined and demonstrated?

Asking the above questions makes explicit how energy planning and design necessitates engagement with broad public interests, including concerns of equity and justice (78). Although DOE made new efforts in this regard, we reflect below on key institutional barriers we experienced first-hand, and present our critical reflections on these barriers as a key result of the effort (3). The critical observations and tone reflect the data of our experience and serve as cautionary procedural results for future efforts to advance CBS. We cover the following four barriers: 1) setting the participatory frames; 2) limiting the scope of discussion, 3) a lack of protection for windows of opportunity with the public, and 4) obstructions to the collection of public input. All of these institutional barriers play out amidst a DOE's culture of secrecy and technocratic decision making (a culture unique among other agencies with and for whom ECAST has conducted pTA activities (57)).

3.2.1 Setting the Participatory Frame

For the DOE's CBS process, the ECAST team set out to design, develop, and assess up to seven geographically distributed pTA forums for collecting informed public views to support CBS process planning and decision making. The 3-step standard ECAST model presented a series of contentions to navigate with DOE, starting with a naming dilemma. The participants in ECAST public forums are selected through an open application process to highlight demographic, geographic, and other diverse attributes of interest. From DOE's perspective, this engagement activity translated to mean a “focus group.” But though application to ECAST forums are open to the public, not all who apply are selected to participate owing to a commitment to the inclusion of key, often underrepresented publics. Also, for ECAST, this term was inappropriate as focus groups do not generally engage in deliberation at the scale of pTA exercises; instead, ECAST compromised with DOE to hold “facilitated representative focus groups,” arriving at a non-existent term ill-suited to the charge of pTAs, but one ultimately amenable to the constraints of DOE for framing participatory engagement. Ultimately, DOE recognized in the pTA forums a potentially valuable complement to prior and parallel public engagement efforts, and ECAST proceeded to plan the “facilitated representative focus groups.”

Rather than an attempt to gain public support for a predetermined outcome like Yucca Mountain, the ECAST pTA approach was to assist in the formation of a legitimate, trustworthy process for public participation in creating more equitable processes for managing SNF. Indeed, the legitimacy of a CBS process relies on the public, especially potential host communities, trusting enough, and being able and willing to invest time into a process where participants feel heard and are respected. The ECAST pTA hoped to contribute to addressing the dearth of public discussions as to the appropriate arrangements of governance and power in technoscientific

concerns and how to reach those arrangements (79) in relation to SNF management. The framing and pre-deliberative conversations and vocabulary of negotiation among experts, stakeholders and decision-makers made the process of engaging the public central, rather than an obstacle, to the DOE's ability to achieve its objective—a critical advancement for improving any relationship with stakeholders or members of the public. However, the challenge of overcoming DOE insistence on position and aggregation rules delimiting what counts as participatory and how, and how to control the corresponding process, surfaced a key tension impacting how publics can be brought into a respected position in an transparent, undetermined process.

3.2.2 Limiting the Scope of Problem

A key feature of ECAST pTA is an explicit, active, and transparent effort to combine expert and citizen experiences, knowledge, and values to inform decision making. Prior to the problem framing workshop with experts and stakeholders, DOE and ECAST met in March 2016 to establish the scope for the pTAs, addressing the question of what would and would not be discussed during the events. DOE insisted the scope of the pTA be limited to the creation of a CBS process. The agency insisted that issues of nuclear energy production and transportation of hazardous waste not be included in deliberations. This demarcation presented a serious design constraint for ECAST, as the nuclear fuel cycle and transportation of nuclear waste are, for certain stakeholders, highly controversial issues in nuclear waste management and potentially rich topics to be able to include in an overarching engagement plan. For DOE, such a scope rule seemed necessary to focus the discussion specifically on the CBS process for a repository and not have its events “derailed” by the concerns of communities and public participants about other aspects of the nuclear cycle.

As noted by expert stakeholders convened by ECAST, dictation of the aspects of SNF management permitted for discussion by DOE was in direct contradiction to the open intent of the CBS approach. Openness toward controversial topics for deliberation—even if the contents are not perceived as directly related to the purpose behind an engagement—is a capacity vital to any future process involving CBS. Such topics need not set the tone for all deliberations, but recognition of their importance as topics of concern to key stakeholders in a CBS process could contribute to creating more trust both through and within these discussions. Determined aversion to controversial topics only further antagonizes interested parties to a CBS process. Breaking and reversing this cycle will be essential to building credibility and legitimacy of any future CBS approach.

3.2.3 Protecting Windows of Political Opportunity

ECAST offered DOE a variety of formats for deliberative pTA processes, including the consensus conference model (80), featuring smaller numbers of individuals meeting over a sustained period (enabling greater depth of deliberation). Fairly early-on, it became clear that the 100-person World Wide Views-style events, which ECAST had supported on numerous occasions (57), had greater appeal to DOE. On reflection, our impression of this preference on the part of DOE was the potential to capitalize on the performativity of large-n events as demonstration of DOE's commitment to broad and inclusive national outreach and engagement. Consequently, ECAST and DOE elected to proceed with simultaneous 100-participants events, representative of local demographic diversities for each public forum, through an open

application process. ECAST planned to hold the day-long public forums concurrently in January or early February 2017 at science museums across the country.

Site selection was based on DOE interest in hosting six to seven forums, with concerns that these: a) not happen where previous recent public meetings had occurred, to avoid appearance of redundancy; b) not be held in states that had communities who had previously expressed interest in hosting SNF repositories sites, so as to avoid any appearance of pre-selection, and c) avoid having conversations dominated by the same groups of public stakeholders consistently present at DOE events. Simultaneity of public deliberations would limit bias by preventing other locations from broadcasting questions or results ahead of time, though all participants would have access to identical background materials for the deliberations. Ultimately, DOE and ECAST agreed on five total sites in the Northeastern, Central, Southern, Northwestern, and Western United States, representing geographic diversity nationally, as well as opportunities to partner with leading science and museum institutions to deliver productive, consistent forum deliberations. This agreement resulted in some redundancies with public meeting locations but addressed DOE concerns of not seeming to broadcast pre-determined site selections.

Although reflecting legitimate, thoughtful concerns, taking additional time to negotiate these details, which initially seemed straightforward based on ECAST's prior projects, resulted in the loss of dynamic windows of opportunity for policymakers to engage with diverse publics. A pattern of espousing a genuine desire to engage the public, which includes the public institutions like science museums, then neglecting to follow-up or cancelling mid-project because of closing political windows, diminishes the credibility of federal agencies and implementing partners and disrespects the very parties such entities claim to want to hear from. The legitimacy of a CBS process relies on engaging the public as a central focus, not an obstacle to CBS. Centering the public in this way, as ECAST efforts demonstrated, is not enough: steps must be taken to insulate a CBS approach from the inconsistent half-lives of political windows. The public needs to be able and willing to invest personal time into a process where they feel heard and are respected. Future processes would benefit from measures to hold such windows open and give procedural-deliberative processes the time and flexibility needed to arrive at such key agreements without fear of having political windows foreclosed prematurely (such as a presidential election).

3.2.4 Legislative Obstructions to the Collection of Public Input

After the expert and stakeholder framing workshop, ECAST partnered with science museums and universities, in consultation with diverse outside experts as well as with the larger CBS policy effort, to design the day-long forums about a CBS process for SNF management. In these forums, participants were to have opportunities to learn and share their perspectives in the form of written responses about nuclear waste management and consent-based siting based on their

interests, knowledge, values, and personal experience.⁶ Forums were to include a combination of informational videos, facilitated conversations, and other participatory activities.

Very early into discussions of data collection methods and instruments, the expedient route of the project soon became a tortuous detour of scheduling delays, methodological compromise, and budget increases. In general, participant data from pTA forums are collected through a) pre-screening survey, b) pre-forum survey, c) in-forum survey, d) table observations, e) post forum surveys, and f) participant interviews. This necessarily includes asking the same questions to hundreds of public participants. The U.S. Paperwork Reduction Act (PRA) of 1995 (PL 104-13)⁷ is triggered when federal projects seek, “Answers to identical questions posed to, or identical reporting or recordkeeping requirements imposed on, ten or more persons, other than agencies, instrumentalities, or employees of the United States.”⁸

For ECAST, the PRA posed several immediate legal and participatory challenges to the standard method of pTA. First, it was unclear whether the statute was applicable to this project; second, which methods of data collection it applied to; and finally, depending on answers to the preceding, ways to achieve objectives through alternative means. Conflicting advice from various general counsels within DOE confused the matter of whether an exception to PRA could be granted for the purpose of the pTA exercise. After several months of back and forth, ECAST was informed that the planned pTA would violate the PRA. As a result, a decision had to be made whether to seek expedited approval from the Office of Management and Budget (OMB) for data collection. ECAST was told OMB approval could take anywhere between 6 to 12 additional months. The approval process to assess burden required public disclosure of all survey questions in advance, which would have compromised the integrity of the ECAST engagement process and utility of several data collection and assessment methods.

ECAST consulted with DOE on alternative data collection methods to maintain the previously agreed timeline for public deliberations. Several possibilities were considered, but the alternative ultimately selected had two components. First, an outside firm would be used for participant recruitment based on demographic criteria agreed upon by ECAST and DOE. This meant that the diversity goals could be addressed only in terms of demographic composition of the selected public, but not in terms of their perspectives and lived experiences, which is counter to ECAST’s normal recruitment process (which involves participants completing a pre-screening survey where non-demographic related questions are asked). Second, all public opinion and rationale data from the deliberation would be collected through observation and group activities instead of written individual responses. Retaining the services of the outside firm and recruiting multiple observers per table added new costs and contributed to further delays as well as a loss of fidelity and utility of the pTA process.

The irony of the ECAST experience here is that DOE originally used a subcontracting mechanism with the network to have the pTA process happen more quickly. At the time, the

⁶ In such an introductory session, for example, participant views on nuclear power could have been explored—not as decision relevant, but rather to showcase for participants the way in which pluralities of values can be expressed and so model the deliberative process of the forum.

⁷ <https://www.congress.gov/bill/104th-congress/senate-bill/244>

⁸ <https://www.reginfo.gov/public/reginfo/prs.pdf>

consideration was that using a standard DOE Request for Proposal (RFP) process would generate unnecessary delays, complications, and administrative reporting burdens. The subsequent discovery of the PRA complications and other issues of the subcontracting mechanism, with its convoluted financial and administrative reporting lines, contributed to generating more problems than solutions. The PRA in this instance legislates a form of ignorance of public concerns that further undermines the ability of the government to learn of issues for and values of publics and communities. Such legislated ignorance seems incompatible with the kind of long-term deliberative effort that a genuine CBS process would necessitate.

4. Conclusions

In January 2017, DOE formally cancelled the ECAST project, citing a shift in executive priorities under a new presidential administration. The swift truncation of the CBS process is emblematic of how the DOE has been struggling with meaningfully engaging public stakeholders for 50 years. The CBS concept is starkly oppositional to the top-down, technocratic approach that has framed previous SNF repository policies. As such, it is not surprising this approach would bring new complexities in structuring public discussions and managing public expectations. ECAST's efforts with pTA to develop a CBS process revealed ways in which the DOE's focus on expediency; limitations on the scope of discussions; bureaucratic issues with collecting public data; lack of continuity across executive administrations; and absence of top level commitment to procedural and institutional learning, innovation, and adaptation, hinder prospects to rectify the technocratic legacies and eroded trust marking the history of U.S. SNF management. In reflecting on these tensions, we hope to smooth the path for future efforts embarking on CBS. Lessons learned, however, could be extended to issues of siting industrial power, waste, and other facilities related to energy systems where legacies of technocratic decision-making and injustice abide (81, 82).

The history of public engagement by DOE around SNF offers insights into how democratic governments erode public trust in federal governance, making it more difficult and fraught for governments to operate transparently, in turn leading to further retrenchment of opposing positions and degradation of trust. One of the key points of the 2012 BRC report was to establish an independent agency to make recommendations and oversee a CBS process (49). Critically, the BRC noted there is no "clean slate" for engaging with the public, as any agency will pull from existing pools of expertise, tempering perceptions of this new agency. Our experience bears out that the institutionalized technocratic culture of DOE is a difficult legacy for a CBS process to overcome. We also observed how restrictions of existing laws, including the NWPA and the PRA, deprive any aspirational CBS process of the time and flexibility necessary to build credibility and legitimacy. From our perspective, open acknowledgement of such difficulties and mistakes across previous site selection processes would contribute to restoring much depleted reserves of trust for any process. Finally, if DOE continuously deprives its constituents of opportunities to make informed decisions over which they have legitimate interest and concern, the agency will invariably return to its ingrained cultural response of trying to "take care of things" through a technocratic process. From our experience, any entity entrusted with stewarding a future CBS approach must take genuine care and responsibility to invest in the capacity of its constituents to make informed decisions. Thus, any CBS process will need to

clarify how communities can obtain and make sense of this information in a manner conforming to their needs and as part of a remedy to the larger injustices done by the power imbalance endemic to technocratic approaches.

The CBS process generally represents a means of responding to the broader contours and specific details of socio-technical governance challenges in democracies. Too often, these are challenges the public is rarely enrolled in or valued to contribute to resolving (83). One benefit of the cancelled pTA is the affordance for policy makers and stakeholders to learn from pitfalls along the path of one such effort to include and value public participation. One can imagine several innovative ways to learn from such experiences, such as creating a training program for DOE program managers on project design and implementation in the context and history of SNF management, within the frame of participatory justice. An additional possible antidote to DOE's reticence and perhaps anxiety at ceding control to more deliberative processes could be to adopt an intentional approach to experimenting with different forms of participatory engagement. The DOE could focus on building its capacity with different formats of public engagement and so contribute to informing a long-term legitimacy of a CBS process rather than focusing on expediency or efficiency in the process (incidentally contributing to building its integrity and public image in the process). These formats could include small, consensus group models over time, or the completion of the ECAST public forums, which even in revised form, could yield useful methods for creating an equitable CBS process. Critically, there is no one method to satisfy participatory justice issues; any process must be flexible and will be time-consuming.

Independent of any CBS process, DOE should engage in such critical reflection with an eye toward relationship healing by focusing efforts on building public trust, rather than technical outcomes. As this manuscript goes to press, DOE has already launched a new request for information on a CBS process for interim storage of SNF, signaling a renewed effort in CBS after a four year hiatus.⁹ If appropriately stipulated and supported, the CBS process offers the U.S. a ladder path out of the technocratic hole of its history with SNF management. The reconciliation on offer and made possible by leveraging the best parts of our democratic institutions to govern this intergenerational science and technology policy challenge demands nothing less.

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